

Course Code	Name of the Course				
216U03E611	Basic VLSI Design				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100
Course pre-requisites: Concepts of Electronic Devices, Digital Circuits, Integrated circuits					
Course Objectives: This course provides students basic knowledge and skills in the fundamental theories and practices of Cyber Security. The course will gives the knowledge and techniques of protecting computers, networks, programs and data from unauthorized access or attacks that are aimed for exploitation.					
Course Outcomes (CO): At the end of successful completion of the course the student will be able to CO1. Demonstrate an understanding of MOS fundamentals CO2. Design and analyse combinational logic circuits CO3. Design and analyse sequential logic circuits CO4. Implement different dynamic and low power CMOS circuit CO5. Implementation Digital circuits using HDL in CPLD/FPGA					

Detailed Syllabus

Module No.	Unit No.	Details	Hrs.	CO
1	Introduction		08	CO 1
	1.1	Introduction, Revolutions in VLSI, Overview of VLSI design flow		
	1.2	Semiconductor physics of MOSFET, MOSFET Types: PMOS, NMOS and CMOS; threshold voltage calculations ;MOSFET capacitances, MOSFET Fabrication		
	1.3	Stick diagram and Layout design (using lamda rules), scaling		
	1.4	Self-Study: NMOS, PMOS and CMOS fabrication process flow and Basic processes		
2	MOSFET combinational circuits		08	CO2
	2.1	Inverter Characteristics, CMOS Inverter Operation, Voltage Transfer Characteristics, Noise Margins and Power consumption		
	2.2	Transmission Gate (TG), Pass Transistor Logic (PTL), Complementary pass transistor Logic (CPL), CPL based logic dates and combinational circuits		
	2.3	Logic circuit Design: Analysis of two input NOR and NAND gate using equivalent CMOS Inverter		
	2.4	Arithmetic building blocks: Adder, ripple carry adder, carry look ahead adder, Multipliers: Array multiplier, Wallace Tree multiplier		
3	Sequential MOS Logic Circuits		08	CO3
	3.1	Bistable circuits, Design for sequential circuits like SR latch, Clocked latch and Flip-Flops , J-K flip flop, D latch and edge triggered Flip-Flop, Sense-Amplifier-Based Flip-Flops, Power consumption of clocking system and power saving methodologies		
	3.3	Semiconductor Memories, memory types, Dynamic RAM, configuration of DRAM, DRAM cell types, Flash Memory		
4	Dynamic Logic Circuits and Low power CMOS circuits		10	CO4
	4.1	Basic Principle of pass Transistor circuits, Voltage Bootstrapping		
	4.2	Synchronous Dynamic circuit techniques, Dynamic Pass Transistor circuits, Dynamic CMOS circuits techniques,		
	4.3	High-Performance Dynamic CMOS Circuits, Domino CMOS Logic, NORA CMOS Logic, Zipper CMOS Logic, TSPC Dynamic CMOS		

	4.4	Overview of power consumption, Low power design through voltage scaling, variable-threshold CMOS circuits, Parallel processing approach		
5	Programmable logic Devices and Hardware Description Languages		10	CO5
	5.1	Architecture of PLDs, PROM (Programmable Read-Only Memory), PAL (Programmable Array Logic), PLA (Programmable Logic Array), Complex PLDs (CPLDs), Field-Programmable Gate Arrays (FPGAs), Implementation of digital circuits in CPLD and FPGA		
	5.2	Types of HDLs, Basic constructs and modelling using VHDL (Very High-Speed Integrated Circuit Hardware Description Language) and Verilog HDL		
	5.3	Digital Circuits Using HDL, Combinational Circuits: Multiplexers, Encoders, Decoders, MUX in Verilog and VHDL. Sequential Circuits: Flip-Flops (D, JK, T), Counters and Shift Registers		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Sung-Mo Kang, Yusuf Leblebici, Chulwoo Kim	<i>CMOS Digital Integrated Circuits</i>	McGraw-Hill Education	4 TH Edition, 2015
2	John P. Uyemura	<i>VLSI Circuits and Systems</i>	Wiley Publications, India	1st Edition, 2001
3	Neil H. E. Weste, Kamran Esheaghian	<i>Principles of CMOS VLSI Design</i>	Pearson Education India,	4th Edition, 2006
4	J. Bhaskar	<i>A VHDL Primer</i>	Pearson	3rd Edition, 2020
5	Samir Palnitkar	<i>Verilog HDL A guide to Digital Design and Synthesis</i>	SunSoft Press	2 nd Edition 1996